

PARTS LIST

Semiconductors:

IC1, IC2	- LM358 op-amp
D1	- 1N4007 rectifier diode
LED1	- 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1, R2	- 100-kilo-ohm
R3, R4	- 220-ohm
R5, R6	- 10-kilo-ohm
R7, R8	- 1-mega-ohm
R9-R12	- 330-ohm
R13-R16	- 33-kilo-ohm
R17	- 3.3-kilo-ohm

Capacitors:

C1, C2	- 0.33 μ F ceramic disk
C3, C4	- 0.22nF ceramic disk
C5-C7, C10	- 100 μ F, 35V electrolytic
C8, C9	- 100nF ceramic disk

Miscellaneous:

CON1	- 2-pin terminal connector
CON2-CON5	- 3-pin connector
CON6, CON7	- 4-pin USB (Type-A) connector
S1	- Toggle switch (SPDT)
SJ1-SJ6	- Shorting jumper
	- 12V DC power supply or
	5V USB power supply
	- Audio source
	- Headphone
	- 8-pin IC socket for IC1 and IC2

pendent channels (inputs 1 and 2): Each channel has a gain of +1. Each headphone is driven by the corre-

sponding amplifier. Make sure to close shorting jumpers SJ1 through SJ4, while leaving SJ5 and SJ6 open.

2. Mono buffer for two headphones: Right channel input is used, while left channel is not used. Buffer channel has gain of +1. The same signal is available on all outputs from the right channel. Make sure to close SJ1, SJ2 and SJ5, and open SJ3, SJ4 and SJ6.

3. Mono buffer for single headphone with signal from the first right channel: In this case, all op-amps drive the same load (single headphone or loudspeaker). Make sure to close SJ1, SJ2, SJ5 and SJ6, and open SJ3 and SJ4.

4. Mono buffer summing both channels in one signal and driving both headphones with the same signal (left and right): For this configuration, close SJ2, SJ4 and SJ5, and open SJ1, SJ3 and SJ6.

Construction and testing

An actual-size PCB layout of the configurable audio buffer for headphones is shown in Fig. 2 and its components

layout in Fig. 3. After assembling the circuit on the PCB, enclose it in a suitable box.

The exact power supply of the amplifier depends on selected load impedance of the headphones, required output power and acceptable level of total harmonic distortion (THD). The most important limitation is to not overheat the op-amps.

Power supply of the circuit can be applied on CON1, or any USB connector (CON6 or CON7). Both USB connectors are in parallel, so if you use the first one, you can still use the second to connect most USB devices.

Equalisation resistors R9 through R12 can be from 15-ohm to around 39-ohm, depending on loads, power supply and op-amps. **EFY**



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